

Python OOP, Decorators & AsyncIO Guide

1. Object-Oriented Programming

Classes, Encapsulation, Inheritance, Polymorphism, Method Overriding, and Dunder methods (`__init__`, `__str__`, `__repr__`, `__add__`).

2. Decorators & Closures

Functions as first-class objects, closures, function wrappers, and `@functools.wraps`.

3. Concurrency & AsyncIO

Threading (I/O bound), Multiprocessing (CPU bound bypassing GIL), and AsyncIO event loops (`async/await`).